

SEMENIC, Romania

WMO: 153150

Lat: 45.1814N

Lon: 22.0558E

Elev: 4702

StdP: 12.36

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.4	(c) 8.6	(d) -7.9	(e) 4.3	(f) 17.9	(g) -0.6	(h) 6.3	(i) 20.3	(j) 35.0	(k) 19.2	(l) 31.7	(m) 22.6	(n) 13.1	(o) 130	(p) 0.729

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.5	72.7	60.2	69.9	59.4	67.3	58.3	62.4	68.9	61.1	67.2	59.8	65.7	9.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.5	93.7	64.8	58.9	88.6	63.5	57.4	83.8	62.4	30.6	69.0	29.6	67.7	28.6	65.8	68.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 32.9	(b) 28.0	(c) 24.5	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 32.9	(b) 28.0	(c) 24.5	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.9	23.3	26.2	29.5	38.5	47.8	55.1	57.5	58.3	50.2	41.9	34.6	27.1
	DBStd	14.56	8.85	7.76	7.90	9.91	6.58	6.39	5.70	6.33	7.82	8.69	8.54	8.30
	HDD50	4194	829	666	636	358	119	25	7	9	95	277	463	710
	HDD65	8814	1294	1086	1101	794	532	299	240	221	445	716	911	1175
	CDD50	880	0	0	0	15	51	179	240	265	102	26	2	0
	CDD65	27	0	0	0	0	0	3	8	13	3	0	0	0
	CDH74	43	0	0	0	0	0	2	12	24	5	0	0	0
	CDH80	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	10.5	10.9	12.9	11.0	11.0	10.4	8.6	8.0	8.7	10.0	10.3	12.8	11.6
Precipitation	PrecAvg	46.3	2.6	2.6	2.4	3.5	5.5	7.1	5.7	4.9	3.6	2.6	2.7	3.0
	PrecMax	66.3	6.9	6.8	7.6	8.6	12.0	21.7	12.2	13.2	10.1	10.2	6.3	5.8
	PrecMin	26.5	0.3	0.1	0.3	0.9	1.0	2.8	1.4	0.8	0.2	0.3	0.0	0.1
	PrecStd	11.2	1.7	1.6	1.4	1.7	2.7	3.9	2.8	3.3	2.4	2.5	1.5	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.6	47.6	51.3	62.5	67.8	72.0	75.5	77.0	73.1	64.6	56.8	49.1
		MCWB	34.5	40.7	43.0	48.7	55.7	61.0	60.9	62.3	60.7	50.8	40.8	37.9
	2%	DB	39.9	41.7	46.1	59.3	62.9	69.5	71.8	73.3	67.2	60.8	51.4	43.9
		MCWB	34.0	37.4	39.2	48.5	53.1	61.1	60.4	59.9	56.6	50.5	39.7	36.3
	5%	DB	37.6	38.6	43.1	55.2	59.9	67.2	68.8	70.2	63.5	57.2	48.1	40.5
		MCWB	34.2	35.1	37.2	45.8	51.4	59.4	58.6	59.3	55.3	47.4	40.8	35.6
	10%	DB	34.7	36.2	39.9	51.9	57.2	64.5	66.0	67.1	60.7	53.9	44.9	37.5
		MCWB	32.4	33.6	35.1	44.5	50.1	57.7	58.1	58.2	53.6	46.9	41.4	33.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.7	42.6	43.9	52.7	56.8	64.7	63.5	65.2	60.6	54.6	46.4	41.9
		MCDB	40.6	45.8	51.9	58.5	63.5	68.6	70.2	71.6	70.5	60.3	49.0	45.6
	2%	WB	37.0	39.1	40.6	49.3	54.6	62.2	61.8	62.2	58.2	52.3	44.9	38.1
		MCDB	38.6	40.8	44.7	56.3	60.8	67.4	68.2	69.2	66.0	58.0	46.8	41.4
	5%	WB	34.4	36.0	38.2	47.3	52.6	60.6	60.5	60.9	55.9	50.0	43.6	36.3
		MCDB	35.7	37.6	42.0	54.2	57.9	65.7	66.5	67.2	61.7	54.9	45.3	39.9
	10%	WB	32.6	34.1	36.1	44.9	50.9	58.7	59.1	59.4	54.3	47.9	42.0	34.6
		MCDB	34.2	35.3	39.1	51.4	56.0	63.8	64.9	65.5	59.2	52.8	44.0	37.3
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.0	9.5	10.9	10.2	10.8	10.7	10.5	9.4	9.8	8.7	8.5
		MCDBR	11.1	11.3	13.8	14.6	13.4	14.0	14.0	13.8	12.8	14.1	12.9	10.4
		MCWBR	8.8	8.1	8.1	8.6	8.3	9.3	9.1	8.2	7.8	9.4	7.8	6.7
	5% WB	MCDBR	9.9	10.4	11.8	13.3	11.7	12.9	12.2	12.6	11.9	12.6	8.4	10.6
		MCWBR	8.5	7.7	8.1	8.1	8.3	9.7	8.8	8.2	7.8	9.5	6.5	8.6
Clear-Sky Solar Irradiance	taub	0.272	0.294	0.324	0.365	0.357	0.351	0.360	0.370	0.346	0.317	0.294	0.273	
	taud	2.430	2.379	2.385	2.288	2.344	2.380	2.348	2.320	2.380	2.473	2.499	2.470	
	Ebn at Noon	271	284	289	284	289	290	286	279	277	272	260	259	
	Edh at Noon	22	29	33	39	38	37	38	38	33	26	21	19	
All-Sky Solar Radiation	RadAvg	403	661	1039	1424	1715	1918	1910	1713	1207	824	489	335	
	RadStd	50	87	139	152	158	177	149	166	125	96	55	47	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)		+1.19	N/A	N/A	N/A	N/A	N/A	-198	-311	+212	+112

Nomenclature: See separate page